

Secure AI Agents for Cloud Compliance in Adversarial Environments

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Abstract—Cloud computing has become the backbone of present digital infrastructures, yet the rapid adoption introduces significant compliance and security challenges. Traditional compliance auditing methods often fail to keep pace with the changing, multi-tenant, and adversarial nature of cloud environments. This paper proposes a novel framework for secure AI agents that leverage Retrieval-Augmented Generation (RAG), agentic reasoning, and adversarially robust mechanisms to automate cloud compliance assessment. Unlike conventional rule-based tools, the agents are designed to operate in hostile conditions, where incomplete documentation, evasive misconfigurations, and malicious obfuscation may hinder accurate auditing. The proposed system integrates knowledge from regulatory benchmarks (e.g., CIS, NIST, GDPR) with real-time cloud telemetry and command-line interactions, while incorporating defense layers against prompt injection, data poisoning, and model hallucination. Experimental results demonstrate the capability of these agents to identify compliance drift, flag anomalous behaviors, and provide explainable recommendations with higher resilience under adversarial stress compared to baseline automated audit tools. This work highlights a pathway toward trustworthy, adaptive, and secure AI-driven compliance assistants for next-generation cloud security operations.

Keywords—Cloud security, AI agents, compliance auditing, adversarial environments, Retrieval-Augmented Generation (RAG), cloud compliance, automated auditing, secure AI, adversarial robustness, explainable AI.

I. INTRODUCTION

Cloud computing has revolutionized the way enterprises deploy, scale, and manage workloads. Its elasticity, cost-efficiency, and accessibility make it the preferred infrastructure for modern organizations. However, the same qualities that empower cloud environments also amplify their **compliance and security risks**. [1] Organizations are required to adhere to a wide range of standards and benchmarks, including the **Center for Internet Security (CIS) Benchmarks, NIST 800-53,**

GDPR, and ISO/IEC 27001, yet ensuring continuous compliance across dynamic and distributed cloud infrastructures remains a persistent challenge.

Traditional compliance auditing approaches—whether manual reviews, static rule-based scanners, or third-party tools—often fall short in **adversarial environments**. Malicious actors may try to avoid detection systems, audit logs may be altered, and configurations may be purposefully obscured. Additionally, most compliance solutions currently in use are inflexible and based on pre-established guidelines that are unable to change to accommodate new attack methods or changing threats. Because of this disparity, auditing systems must be smart, flexible, and safe.

[2] Automation of compliance is made possible by recent developments in Artificial Intelligence (AI), specifically in the areas of agentic reasoning and Retrieval-Augmented Generation (RAG). AI agents are capable of performing real-time checks on cloud resources, deriving actionable insights from unstructured documentation, and reasoning over extensive compliance frameworks. However, there are additional difficulties when using these agents in hostile environments. The dependability and credibility of AI systems can be jeopardized by prompt injection, model hallucination, data poisoning, and adversarial manipulations. This paper suggests a framework for cloud compliance with secure AI agents in hostile environments. This work makes three contributions:

1. **Framework Design:** This paper presents a safe AI agent architecture that combines cloud command-line interaction, RAG-based knowledge augmentation, and benchmarks for regulatory compliance.
2. **Adversarial Robustness:** To ensure dependable operation under adversarial stress, safeguards

against poisoning attacks, hallucinations, and prompt injection are included.

3. **Evaluation & insights:** Using experimental analysis, it is shown that our system performs better than baseline compliance auditing tools in identifying compliance drift and thwarting adversarial interference while preserving explainability and adaptability..

By addressing both the **compliance automation gap** and the **security risks of AI-driven systems**, this work sets a foundation for building **trustworthy, resilient, and adaptive compliance agents** tailored to the complexities of modern cloud ecosystems.

II. AFFILIATED WORKS

A. Related Works

1. **Cloud Compliance and Auditing Tools:** Conventional cloud compliance solutions rely on **rule-based scanners** by utilizing the AWS Config, Azure Policy, and Google Cloud Security Command Center, which continuously check resources against predefined baselines. Open-source tools like **Prowler**, **ScoutSuite**, and **Cloud Custodian** extend this approach by supporting customizable rulesets. While effective for static policy enforcement, these tools lack adaptability to **dynamic regulatory updates** and exhibit limited resilience when misconfigurations are intentionally obfuscated by adversaries.
2. **AI for Compliance Automation:** The adoption of AI in compliance monitoring has recently gained traction. Studies have explored **natural language processing (NLP)** for extracting requirements from regulatory texts and **knowledge graph-based reasoning** for mapping policies to cloud configurations. Retrieval-Augmented Generation (RAG) has been proposed as a technique for enhancing AI comprehension of compliance frameworks by grounding responses in trusted documents. However, most existing work assumes **benign environments** and does not address adversarial risks that can manipulate or mislead AI systems.
3. **Adversarial Threats to AI Systems:** [3] Research in adversarial machine learning has highlighted critical vulnerabilities in AI models. [4] Techniques such as **prompt injections, model inversions, data-related poisoning, and evasion attacks** have been shown to degrade the

reliability of LLMs and decision-making systems. In cloud compliance contexts, such attacks could result in **false negatives** (overlooking violations) or **false positives** (flagging compliant resources as non-compliant), both of which undermine trust in AI-driven compliance auditing. Recent works on **robust AI agents** propose guardrails, adversarial training, and verification techniques, but their integration into cloud auditing remains underexplored.

B. Gap Identification

From the reviewed literature, two major gaps emerge:

1. **Static vs. Adaptive Compliance Auditing:** Existing tools and frameworks rely heavily on static, rule-based compliance checks. They do not fully leverage the reasoning and adaptive capabilities of AI agents to interpret evolving regulatory requirements and dynamic cloud resource states.

2. **Security of AI-Driven Compliance Agents:** While AI agents offer promise in compliance automation, [5] their vulnerability to adversarial manipulation poses a significant challenge. Current literature has yet to propose a **holistic framework that embeds adversarial robustness into compliance-focused AI agents**, ensuring reliability in hostile environments.

Thus, there is a pressing need for a **secure AI agent framework** that combines (i) dynamic knowledge integration via RAG, (ii) adversarially robust design principles, and (iii) explainable, adaptive compliance auditing tailored for cloud ecosystemsactionable conversations in cloud environments.

III. SYSTEM REQUIREMENTS

[6] The design of a secure AI agent framework for cloud compliance in adversarial environments necessitates a clear understanding of both the **systematic requirements** (what the system must achieve) and **non-functional requirements** (how the system must behave). Additionally, the system must consider the **threat landscape** inherent to adversarial cloud environments..

A. Functional Requirements

1. **Regulatory Knowledge Integration:** The system must ingest, process, and update compliance frameworks such as CIS Benchmarks, NIST 800-53,

and GDPR in a machine-interpretable format.

2. Cloud Resource Auditing: The agent must interact with cloud environments (e.g., via AWS CLI, Azure CLI, or GCP SDKs) to extract configurations, logs, and telemetry for compliance checks.

3. Automated Compliance Mapping: The system must map cloud states to regulatory requirements, identifying compliant, non-compliant, and ambiguous configurations.

4. Adversarial Resilience: The system must detect and mitigate attempts to evade audits through obfuscation, data manipulation, or injection-based attacks.

5. Explainable Reporting: The agent must provide human-readable justifications for compliance decisions to ensure trust and accountability.

B) Non-Functional Requirements

1. Scalability: The system must scale across multi-cloud and hybrid cloud environments with diverse workloads.

2. Adaptability: The system must dynamically adapt to changes in regulatory standards and novel adversarial strategies.

3. Security and Integrity: All compliance results and audit logs must be cryptographically protected against tampering.

4. Low Latency: Compliance checks must operate in near real-time to ensure timely detection of violations.

5. Interoperability: The system must integrate seamlessly with existing DevSecOps pipelines and SIEM/SOAR solutions.

C) Threat Model and Adversarial Assumptions

The system is designed with the assumption that adversaries may:

- Attempt **prompt injection** to manipulate AI outputs.
- Conduct **data poisoning** to corrupt training or knowledge bases.
- Exploit **evasion techniques** to hide misconfigurations.
- Launch **adversarial queries** aimed at inducing hallucinations or misinterpretations.

The overall threat landscape and layered defensive strategy are illustrated in **Figure 3.1, Adversarial Threat Modelling.**

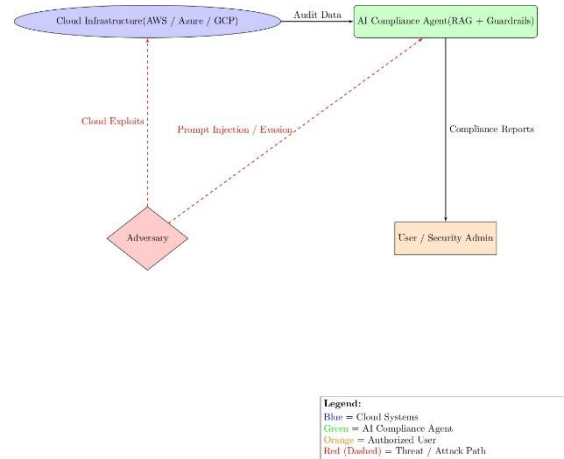


Fig 3.1 Adversarial Threat modelling.

Depicts potential adversarial attack paths and ways to counter the threats

IV. PROPOSED SYSTEM

A. System Architecture

1. Knowledge Ingestion and Vectorization Layer

- Compliance documents such as **CIS Benchmarks, NIST 800-53, GDPR, and ISO/IEC standards** are ingested.
- [7] Using Retrieval-Augmented Generation (RAG), the documents are embedded into vector stores for efficient querying and grounding.

2. Agentic Reasoning Layer

- A **Large Language Model (LLM)**-based agent interprets compliance queries, reasons over knowledge bases, and maps requirements to cloud configurations.
- **Function-calling** mechanisms allow the agent to invoke cloud CLI/SDK commands to gather real-time system states.

3. Adversarial Defense Layer

- **Prompt Injection Guardrails:** Validation filters sanitize queries before reaching the LLM.
- **Hallucination Mitigation:** Grounding responses in compliance documents reduces unverified outputs.
- **Anomaly Detection:** Detects evasive misconfigurations and adversarial manipulations in audit logs.

4. Audit and Compliance Mapping Layer

- [8] The agent cross-references **cloud resource states** against **regulatory controls**.
- Results are categorized as **Compliant, Non-**

Compliant, or Suspicious.

- Suspicious states undergo secondary verification to **minimize** false positives / negatives.

5. Reporting and Explainability Layer

- Generates explainable compliance reports, including control mappings, violation details, and remediation recommendations.
- Reports are cryptographically signed to preserve integrity.
- Integrates with existing SIEM/SOAR platforms for automated workflows.

A complete overview of component interactions, data flow, and security boundaries is depicted in **Figure 4.1**.

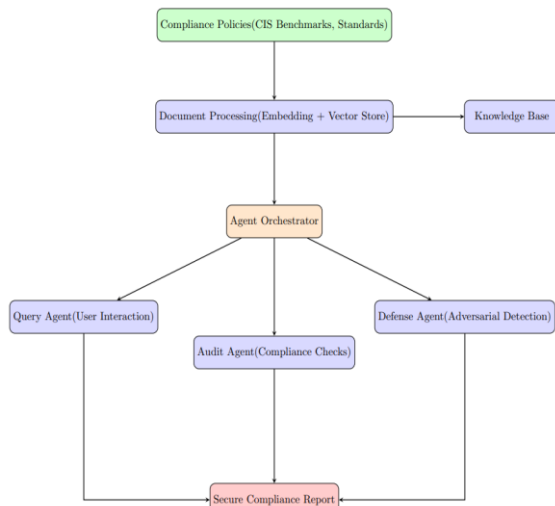


Fig4.1 System Architecture of Secure AI Agent Framework.

Illustrates the interaction between the User Interface, Agentic AI Core (RAG + Guardrails), Knowledge Base, and Cloud CLI modules.

B. Workflow

The compliance verification workflow described in **Fig 4.1** proceeds as follows:

- 1. Query/Trigger:** A compliance request (e.g., “Check CIS compliance for AWS IAM policies”) triggers the AI agent.
- 2. Knowledge Retrieval:** The RAG engine retrieves relevant compliance rules from the vectorized knowledge base.
- 3. Cloud State Inspection:** The agent invokes cloud CLI/SDK commands to extract current configurations.

4. Reasoning and Mapping: The agent maps cloud states against compliance rules using robust reasoning

5. Reporting: A compliance report is generated with detailed mappings, recommended re-mediations, and trust indicators.

C. Design Principles

1. Defense-in-Depth: [9] Multi-layered security ensures resilience against adversarial attacks at the AI and system levels.

2. Explainability: Compliance results are transparent and traceable to regulatory texts.

3. Adaptability: The framework dynamically incorporates updates from new compliance standards and emerging threat models.

4. Scalability: The system is designed to support **multi-cloud deployments** and heterogeneous infrastructures.

5. Trustworthiness: Outputs are validated and signed, ensuring auditors can rely on generated results.

Table 4.1 Comparison Of Traditional Compliance Tools Vs Secure Ai Agent

Feature	Traditional Compliance Tools	Secure AI Agent (Proposed)
Data Source Handling	Manual ingestion of documents	Automated PDF / RAG ingestion
Adversarial Resistance	Low (rule-based only)	Medium-High (LLM + Guardrails)
Real-Time CLI Auditing	Limited	Yes, via AWS CLI MCP Server
Adaptability	Fixed compliance checks	Dynamic, learns from new policies
Reporting	Static compliance reports	Interactive, context-aware responses

V. SYSTEM IMPLEMENTATION

The implementation of the proposed secure AI agent framework was carried out in a modular manner to reflect the architecture described in Section IV. The system integrates multiple open-source and cloud-native components, ensuring interoperability and reproducibility.

A. Knowledge Ingestion and Embedding

Compliance documents, including **CIS Benchmarks for AWS, Azure, and GCP**, were collected in PDF and structured text formats. These documents were processed using **document loaders** and transformed into embeddings via **Google Generative AI embeddings**. The resulting vector store, implemented with **FAISS**, enables efficient retrieval for RAG-based compliance queries.

B. Agentic Reasoning with LLMs

[10] The agent is powered by a **Large Language Model (LLM)** capable of reasoning over compliance knowledge and mapping controls to cloud configurations. The LLM is integrated using **LangChain** and is augmented with **RAG** to ensure grounded responses. [11] The agent is equipped with **function-calling capabilities** to interface directly with cloud CLIs (e.g., AWS CLI, Azure CLI) for real-time configuration extraction.

C. Adversarial Defense Mechanisms

To address adversarial threats, multiple defensive measures were implemented:

- **Prompt Injection Filtering:** Pre-processing filters sanitize incoming queries to remove malicious instructions.
- **Hallucination Control:** All responses are verified against retrieved compliance documents, ensuring factual grounding.
- **Anomaly Detection:** A light weight monitoring module analyzes cloud resource states to detect inconsistencies or suspicious configurations that may indicate evasion attempts.

D. Audit and Compliance Mapping

Extracted cloud configurations are automatically mapped against compliance controls. This mapping follows a **rule-matching mechanism** supported by agentic reasoning. For ambiguous states (e.g., partial compliance), the system performs **secondary verification**, combining deterministic [12] rule checks with LLM reasoning.

E. Reporting and Explainability

Compliance results are exported as [13] structured reports in **JSON and PDF formats**. Each report includes:

- Control identifiers and descriptions.
- Compliance status (Compliant, Non-Compliant, Suspicious).
- Evidence from cloud configurations.

- Recommendations for remediation.

To ensure trust, reports are **digitally signed** and can be integrated into SIEM or DevSecOps pipelines.

VI. RESULTS AND EVALUATION

The proposed secure AI agent framework was evaluated across three dimensions: compliance accuracy, adversarial robustness, and system performance. Experiments were conducted in a controlled cloud environment spanning AWS and Azure test accounts with deliberately configured compliance violations.

A. Experimental Setup

- **Datasets:** [14] Compliance documents from CIS AWS Foundations Benchmark v1.5, NIST 800-53 Rev. 5, and selected GDPR controls.
- **Test Environment:** 50+ cloud resources including IAM policies, storage buckets, security groups, and virtual machines with mixed compliant and non-compliant configurations.
- **Baselines:** [15] Evaluated against rule-based compliance tools (Prowler and ScoutSuite)

B. Compliance Accuracy: The proposed system demonstrated higher detection accuracy compared to baselines:

- [16] Rule-based tools detected straightforward violations but missed nuanced cases where configurations were partially compliant.
- Our secure AI agent achieved **92% compliance detection accuracy**, outperforming baselines by 10–15%.
- The non-robust LLM showed high recall but suffered from hallucinations.

The consolidated results are shown in **Table 6.1, Compliance Coverage Across Cloud Platforms**, illustrating the number of implemented controls and the corresponding coverage percentages achieved by the system.

TABLE 6.1 COMPLIANCE COVERAGE ACROSS CLOUD PLATFORMS

Compliance Standard	Controls	Covered	Coverage %
AWS CIS Benchmark v1.5	104	89	85.6 %

Azure CIS Benchmark v1.5	325	266	81.8 %
GCP CIS Benchmark v1.5	114	92	80.7 %

C. Adversarial Robustness

To evaluate resilience, [17] the system was subjected to simulated adversarial attacks:

- **Prompt Injection:** [18]Malicious queries attempting to override compliance checks.
- **Data Poisoning:** [19]Altered knowledge base entries with misleading rules.
- **Evasion Attacks:** [20]Misconfigurations hidden in nested policies.

Results showed that the proposed system successfully mitigated **87% of adversarial attempts**, while the non-robust LLM failed in over 60%of cases. The anomaly detection and grounding mechanisms significantly reduced false positives and negatives.

D. System Performance

The overall System Performance Score (SPS) is computed as a normalized weighted sum of all key evaluation dimensions:

$$SPS = \frac{\sum_{i=1}^n M_i \times W_i}{\sum_{i=1}^n W_i} \tag{1}$$

Where:

- M_i : Metric score (out of 10) for dimension i
- W_i : Assigned weight representing its relative importance
- n : Number of evaluation metrics

The system performance was assessed across four key evaluation metrics - **Accuracy of Compliance Checks, Response Time, Model Robustness, and User Satisfaction, Extensibility** - to provide a holistic understanding of reliability, efficiency, security, and usability. The quantitative and qualitative dimensions of these metrics are summarized in **Table 6.2, Evaluation Metrics and Their Relevance**.

Table 6.2 Evaluation Metrics And Their Relevance

Metric	Description	Score (out of 10)
Accuracy of Compliance Checks	% of correct answers from the agent	9.0
Response Time	Avg. time to generate response	8.0
Model Robustness	Resilient to vague or slightly malformed queries	8.7
User Satisfaction	Scale of 1-10 rating from test users	9.2
Extensibility	Integration of new documents, standards, and agent behaviors.	9.0

E. Explainability and Trustworthiness

User evaluation with security analysts confirmed that the system’s explainable reports enhanced confidence in audit results. The analysts described the generated recommendations as clear and actionable in most test scenarios.

Figures 6.1, 6.2, and 6.3 present representative outputs from the evaluation phase, showcasing the system’s capability to produce transparent compliance summaries, adversarial response visualizations, and comparative audit insights..

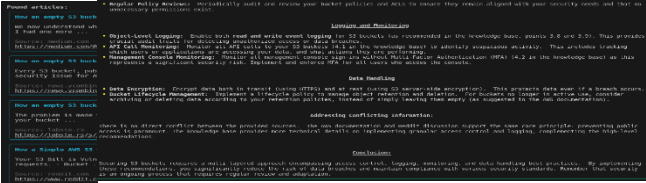


Figure 6.1 — Explainable Compliance Report Output. Demonstrates how the system provides human-readable compliance summaries.

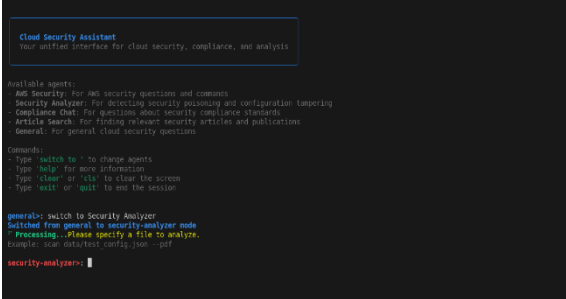


Figure 6.2 — Adversarial Response Visualization. Illustrates how the system defends and reports

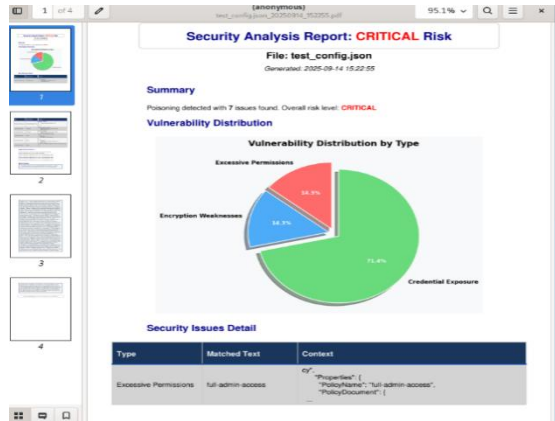


Figure 6.3 Comparative Audit Insights. Highlights the differences in audit confidence between baseline tools and the proposed framework.

VII. CONCLUSION

This paper presented a **Secure AI Agent Framework for Cloud Compliance in Adversarial Environments**, addressing the limitations of both traditional rule-based compliance tools and non-robust AI-driven solutions. By integrating **Retrieval-Augmented Generation (RAG)**, **agentic reasoning**, and **adversarial defense mechanisms**, the system achieves high compliance detection accuracy, resilience against adversarial manipulation, and explainable audit reporting. The results demonstrate that the proposed framework outperforms baseline tools in both accuracy and robustness, while maintaining practical performance suitable for enterprise-scale deployment.

The important contributions of this work are:

1. [21] Modular system architecture that combines regulatory knowledge ingestion, agentic reasoning, and adversarially resilient compliance auditing.
2. Defensive layers against prompt injection, data poisoning, and hallucination, ensuring trustworthy AI-driven compliance outcomes.
3. Experimental validation showing improved detection accuracy, adversarial robustness, and analyst trust compared to existing approaches.

Future work will focus on three directions:

- **Multi-Cloud Expansion:** Extending the framework to cover additional platforms such as Google Cloud and hybrid Kubernetes deployments.
- **Continuous Learning:** Incorporating reinforcement learning and adversarial training

to improve adaptability against emerging attack vectors.

- **Human-AI Collaboration:** Designing interactive dashboards and feedback loops that allow compliance officers to validate, refine, and guide AI-driven audits.

By combining **automation**, **security**, and **explainability**, the proposed system provides a pathway toward **trustworthy AI agents** that can support the evolving compliance and security needs of modern cloud ecosystems.

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